



ABILAKSHAN VADIVEL

UNDERGRADUATE

B.Sc.(Hons) in Surveying Sciences, Specializing in GIS

CONTACT

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LinkedIn GitHub

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Jaffna, Sri Lanka

SURVEYING SKILLS & COMPETENCIES

FIELD SURVEYING

- Topographic Surveying (RTK)
- Engineering Surveying
- Control Point Establishment (Static GPS)
- Boundary Surveys
- Road Design Surveys
- Construction Layout.

GEOSPATIAL ANALYSIS

- Spatial Statistics
- Terrain Modeling (DEM/DTM)
- Vulnerability Mapping (AHP)
- Geostatistical Analysis
- GEOAI
- Digitization
- Data integration
- Visualization
- Spatial queries

SURVEYING INSTRUMENTS

- GNSS/GPS Receivers
- Total Station
- Auto Level & Theodolite
- Drones: DJI Matrice 300 RTK (LiDAR & RGB), DJI Mavic 3 Enterprise (Multispectral)
- Terrestrial Laser Scanner (TLS)
- Ground Penetrating Radar (GPR)

PROFILE

Motivated Surveying Sciences graduate with GIS specialization and practical experience in land demarcation, GNSS-based field support, spatial data preparation, and survey-related mapping workflows. Experienced in CAD-to-GIS conversion, land information system development, thematic map production, and geospatial database preparation using both Esri and open-source GIS tools. Bringing a combined Surveyor and GIS Analyst profile, with the ability to support both field surveying operations and office-based GIS, mapping, and data management tasks.

EXPERIENCE

Planning Intern | Riyan Pvt Ltd, Maldives | December 2025 – March 2026

- Developed LIS workflows using ArcGIS Pro, ArcGIS Online, QGIS, PostgreSQL/PostGIS, and GeoServer.
- Supported CAD-to-GIS conversion, land-use mapping, web GIS preparation, and WMS/WFS publishing.
- Assisted land demarcation, GNSS field support, and survey documentation.
- Contributed to coastal sediment analysis, heat map production, harbor digitization, and drone-data exposure.

GeoAI for Disaster Resilience - Undergraduate Fellowship | ADRIMP Feb 2026 - Present

Management Assistant (IT) - Internship (6 Months)

- Institute of Medical Sciences - Manipay (March - August 2021)
- Provided IT support and managed databases to ensure smooth operations.

EDUCATION

B.Sc. (Hons) in Surveying Sciences Specializing in GIS

Sabaragamuwa University of Sri Lanka - Completed academic requirements
(Up to 3rd year 2nd Semester, GPA: 3.4/4.0)

Full Stack Web Development - 2020 Jun - Nov (ReactJS, NodeJS, MongoDB)

Uki Coding School - Online Cohort 1st Batch

GCE A/L (Physical Science)

St. John's College, Jaffna (2019, 2B, 1C)

PRACTICAL SURVEYING EXPERIENCE & KEY PROJECTS

Topographic Survey & Control Establishment | North Axis (Pvt) Ltd

- Participated in topographic survey for a water tank and sump project in Meesalai, Jaffna.
- Established control points and collected field data for contour plan preparation.

Engineering Survey for Reservoir Construction | Academic Field Project

- Conducted engineering and topographic surveys for a proposed reservoir and access road design.

Survey Camp | Karagasthalawa Maha Vidyalaya, Belihuloya

- Completed topographic and boundary surveys, established GPS control, and prepared site plans.

Advanced Surveying Technology Workshops

- Gained hands-on exposure to LiDAR/multispectral drones, TLS, and GPR.

SURVEYING & GIS SOFTWARE

- AutoCAD & Civil3D
- ArcGIS Pro & ArcGIS Online
- QGIS
- ERDAS Imagine
- Cloud Compare
- Google Earth Engine
- GeoServer
- PgAdmin
- Docker
- GitHub

PROGRAMMING

- Python (Basic), R (Basic), PostgreSQL, HTML/CSS, JavaScript (Basic).

OTHER TOOLS

- Visual Studio Code
- MS Office

LANGUAGES

- Tamil
- English
- Sinhala

MEMBERS

- Royal Institution of Chartered Surveyors (RICS) - Professional Number(Student): 0994438
- Geo Mappers
- ISPRS Student Consortium

ACTIVITIES

Sabaragamuwa University of Sri Lanka

- Table -Tennis team (2022 - 2025)
- Badminton team (2022 - Present)

J/St.Johns' College

- Cricket Team (2012-2018)
- Badminton team (2012-2018)
- Volley Ball Team (2017-2018)
- Table-Tennis team (2012-2016)

INTERESTS

- Gardening
- Drone Videography
- Outdoor and Indoor games.
- Travelling
- Swimming
- Adventures

GIS & ACADEMIC PROJECTS

GIS & Data Visualization Projects

- Esri Story Map: Created an interactive story map to visualize the impacts of the 2019-2020 Australian bushfires.
- Esri Dashboard: Developed a dynamic weather dashboard for real-time data display.
- Geospatial Web & Desktop Applications: Developed a web application for geospatial data visualization using PHP and JavaScript, including a custom GUI for Sri Lanka using Leaflet.js. Also designed a desktop GUI using Python (Tkinter).

Spatial Analysis Projects

- Landslide Vulnerability Mapping: Created a comprehensive landslide vulnerability map for the Matale District using GIS-based spatial analysis and the Analytical Hierarchy Process (AHP) method.
- Geostatistical Analysis in R: Mapped the spatial distribution of coal ash content using geostatistical methods in R.
- Digital Elevation Model (DEM) Comparison: Generated and compared DEMs from digitized contours with SRTM and ASTER data to analyze accuracy and suitability for terrain modeling.

FINAL YEAR RESEARCH

Crowdsourced Data Analysis and Spatial Feature Detection using GeoAI (Australian Bushfire, 2019-2020)

- Developed a GeoAI-based methodology to detect bushfire impacts on features by integrating geotagged crowdsourced data (Tweet - NLP) with multi-temporal Sentinel-2 imagery. Modeled satellite imagery using DeepLabV3+ for automated detection and disaster damage assessment

ACHIEVEMENTS

- Inter-University Badminton Team Vice-Captain - 2025
- Inter-University Table-tennis Runners-up - 2022
- School Badminton team Captain - 2017 & Volley Ball team Captain - 2018
- 2016-2018 School Prefect
- 2017 - Best Cricket Team In Northern Province
- All Island School Cricket under 13 Runners-Up - 2013

CERTIFICATES

ESRI

- Spatial Data Science: The New Frontier in Analytics - MOOC
- Make an Impact with Modern Geo Apps - MOOC
- Going Places with Spatial Analysis - MOOC
- GIS for Climate Action - MOOC
- Cartography - MOOC
- Telling Stories with GIS Maps
- Creating an ArcGIS StoryMaps Briefing
- Introduction to Regression Analysis Using ArcGIS Pro
- Regression Analysis: Building a Regression Model Using ArcGIS Pro

Center for Open & Distance Learning University of Moratuwa, Sri Lanka

- Python for Beginners
- Foundations of Project Management

Prof. RMKGSPB Koswatte

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Character References